Computational Abstractions · LLM Grounding · GBIF Data Publishing

Using AI in Data Publishing

From unstructured spreadsheets to standardized GBIF datasets + hands-on AI exercise.

Begin Lecture →

Open Science Course

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Computational Science in a Nutshell

Modern science is fundamentally computational. Translating empirical observations into digital evidence requires navigating **layers of computational abstraction**:

⚡ The Physical Reality (Silicon)

Computers do not understand species, coordinates, or biodiversity. At the physical layer, CPUs execute **binary voltage states** (`0` and `1`) switching billions of microscopic transistors every second.

`Machine Code · Hardware Opcodes · Registers`

🧠 The Scientific Reality (Mind)

Scientists think in **domain semantics**: *"A specimen of Betula pubescens was collected at 60.1°N with 95% sequencing confidence."* Expressing this requires translating natural concepts into symbols.

`Hypotheses · Taxa · Geospatial Polygons · Provenance`



The Hierarchy of Languages: A Stepping Stone of Abstractions

To bridge silicon and science, computer scientists built progressive tiers of languages — from **Bytecode** to **Assembly**, **C/Rust**, and **Python/R**.

Level 0 · Machine Code & Bytecode

The only language physical hardware actually executes: raw binary opcodes and electrical voltage states.

⚡ What Actually Runs on Silicon

The Central Processing Unit (CPU) contains billions of microscopic transistors acting as physical switches. Instructions are fetched, decoded, and executed strictly as sequences of **bytes and binary bits**.

🔒 **Architecture-Locked:** Machine code compiled for x86-64 cannot execute on ARM64 or Apple Silicon without binary translation.

🧩 **Zero Semantic Meaning:** Raw numbers carry no inherent context — a byte could represent an opcode, an integer, an ASCII letter, or a RAM address.

RAW BINARY / HEX DUMP — x86_64 Opcodes

```
; Raw instructions in memory
48 89 5c 24 08 ; mov [rsp+8], rbx
48 83 ec 20 ; sub rsp, 32
8b 47 04 ; mov eax, [rdi+4]
03 07 ; add eax, [rdi]
c3 ; ret
```

Semantic Distance: **Maximum (Incomprehensible to humans)**

Level 1 · Assembly Language (ASM)

The first step up: replacing raw binary bytes with human-readable symbolic mnemonics.

Direct Hardware Mnemonics

Assembly provides a **1-to-1 symbolic representation** of CPU machine instructions. Instead of remembering binary opcodes like `0x03`, the programmer writes human-readable mnemonics like `ADD`.

Thinking in CPU Registers: The scientist must explicitly manage hardware registers (RAX, RBX, RSP) and memory stack offsets.

No Scientific Abstractions: No dataframes, variables, loops, or types exist — only raw memory addresses, pointers, and arithmetic jumps.

ASSEMBLY LISTING — x86_64 ASM

```
calculate_sum:
  push rbp
  mov rbp, rsp
  mov eax, [rdi] ; load val A
  add eax, [rsi] ; add val B
  pop rbp
  ret
```

Semantic Distance: **Very High (Tied to silicon registers)**

Level 2 · Systems Programming (C & Rust)

Structured programming, manual memory layouts, and compiling down to native binaries.

🦀 Structured Hardware Control

Systems languages introduce **functions, structured types (`struct`), and loops**. They provide deterministic control over memory allocation and CPU cache alignment while compiling directly to native machine code.

📍 **Explicit Pointers & Memory:** Scientists must manually manage heap memory allocations (malloc/free in C, strict borrow lifetimes in Rust).

🚀 **Maximum Computational Speed:** Powers the high-performance engines under our scientific tools (BLAST aligners, GIS engines, matrix backends).

RUST / C STRUCT — Memory Layout

```
struct Occurrence {
  id: u64,
  lat: f64,
  lon: f64,
  taxon_id: u32,
}
// Exactly 24 bytes in memory
```

Semantic Distance: **Medium (Structured, but hardware-bound)**

Level 3 · Scientific Computing in R

Automated memory, domain objects (DataFrames/tibbles), and expressiveness tailored to data analysis.

📊 Expressing Scientific Logic in R

High-level languages like **R** manage memory automatically via garbage collection. Instead of writing byte pointers and managing registers, scientists manipulate **DataFrames, tibbles, vectors, and domain models**.

📋 **Domain-First Data Structures:** DataFrames and tibbles directly mirror how biodiversity researchers organize field observations and experimental spreadsheets.

📦 **Rich Open Science Ecosystem:** Deep ecosystem of specialized R packages (tidyverse, rgbif, sf, vegan) for geospatial mapping, Darwin Core validation, and modeling.

R / TIDYVERSE — DataFrames & Pipes

```r
# Clean & summarize occurrences
clean_occ ← occurrences %>%
  filter(!is.na(decimalLatitude)) %>%
  filter(year ⩾ 2000) %>%
  group_by(scientificName) %>%
  summarise(
    n_obs = n(),
    min_lat = min(decimalLatitude)
  )
```

Semantic Distance — **Low (Speaks in tabular domain objects)**



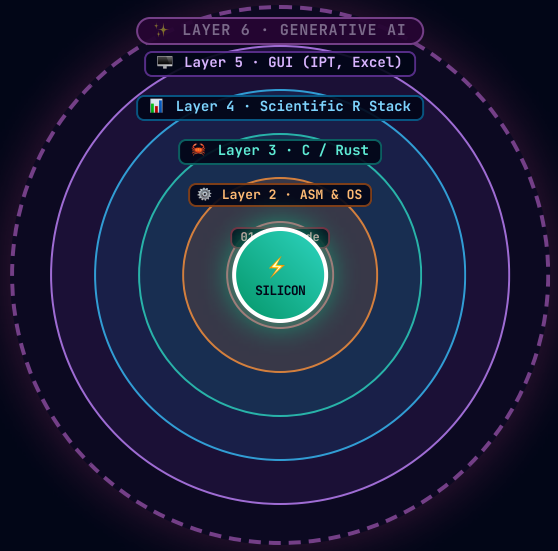
The AI Paradigm Shift: For 70 years, scientists had to learn programming syntax or rely on rigid GUIs. Generative AI collapses semantic distance — the computer now understands **direct human domain intent**.

NEXT: AI IN DATA PUBLISHING →

THE COST OF ABSTRACTION

The Onion of Implicit Trust: Distance from Silicon

Deterministic Physics → Code Trust → Probabilistic Faith



Concentric Abstraction Shells: **Outermost = Maximum Trust Deficit**

Core · Silicon Hardware & Physics

Zero Software Trust

100% Deterministic

Grounded in physical laws: transistor voltage states. Deterministic and verifiable with an oscilloscope. Hardware cannot lie about physical state.

Physical Reality

Layers 1–3 · OS Kernel, ASM & Compilers (C/Rust)

Low-Moderate Trust

Formal Logic

Trusting CPU microcode, OS virtual memory (MMU), process schedulers, and compiler optimization passes (`-O3`) not to alter logic.

Determinism Preserved

Layers 4–5 · R Packages (`tidyverse`) & GUI Apps (IPT, Excel)

Substantial Trust

Supply-Chain Trust

Trusting 50+ nested open-source CRAN package authors, R's garbage collector, and Excel's silent defaults (which corrupted ~20% of published genetic papers!).

Unvetted Codebases

Layer 6 · Generative AI & Large Language Models

MAXIMUM BLIND TRUST

Probabilistic Faith

First non-deterministic layer in human history: Trillions of uninterpretable weights, stochastic sampling, and plausible hallucinations (fake taxa, fabricated DwC terms). Zero formal proofs.

Requires Verification

The Abstraction Paradox: As semantic distance drops to zero, the requirement for human verification skyrockets to 100%. Because AI requires **the most implicit trust of all**, AI in scientific data publishing must always be paired with **active human-in-the-loop validation**.

HUMAN-IN-THE-LOOP MANDATE →

FOUNDATIONAL AI MECHANICS

The Pure LLM: A Next-Token Prediction Engine

No Search · No DB · No Logic Rules · **Only Statistical Probability**

Input Context Window (Tokens in Memory) — Length: t = 10 tokens

In	GBIF	taxonomy	,	the	scientific	name	of	badger	is	Meles ?

Softmax Distribution: P(Token | Context) — Vocabulary: ~100k tokens

✓ "Meles"	■■■■■■■■■■■■■■■■■■■■	89.2%
"Taxidea"	■	6.4%
"Arctonyx"		2.8%
"Mellivora"		1.1%

Sampling: **Temperature = 0.2** (Greedy / Low entropy) — **Token "Meles" sampled**

1 Pure Statistical Pattern Matching — Token Co-occurrence

The model does not know what a badger is, what DNA is, or what biology means. It only knows that in human text, the token `"Meles"` correlates with `"badger"` across millions of training documents.

2 No Database Lookup & No Truth Sensor — Zero Knowledge Base

A pure LLM cannot query SQL, check GBIF, or calculate coordinates. All "knowledge" is frozen in static neural weights. It optimizes for **linguistic plausibility**, not empirical truth.

Why Pure LLMs Hallucinate by Design — CRITICAL SCIENTIFIC LIMIT

If asked for a rare orchid species in a specific county, the LLM will generate Latin-sounding binomials and plausible decimal coordinates. To the model, a *real species* and a *plausible hallucinated species* have identical mathematical status!

Scientific Implication: Grammatical Fluency ≠ Truth

Pure LLMs produce **impeccable grammar and scientific tone** even when producing 100% false Darwin Core mappings. Fluency is an illusion of competence.

The Autoregressive Loop: The sampled token is appended to the prompt, and the *entire sequence* is fed back into the model to predict the next token (`"meles"`), continuing until an `<EOS>` token is reached.

The Fundamental Rule: A pure LLM is an *intuitive author*, not a *verification database*. In biodiversity informatics, never trust an unassisted model to "remember" taxon keys, coordinates, or Darwin Core terms without external tools.

NEXT: AUGMENTING WITH TOOLS →

TAMING THE BEAST · STEP 1

Taming the Beast: Grounding with Authoritative Tools

Unassisted = Hallucination → Tool-Assisted = Anchored Truth

1. Unstructured User Prompt — Mental Domain Intent

```
"Record found: Eurasian badger near Sognsvann, Oslo on
14/05/2023. Map to DwC."
```

⚡ 2. LLM Pauses Generation & Emits Structured Tool Call — No Guesswork

```
// LLM recognizes it needs authoritative ground truth:
tool: gbif_match_taxonomy({
  name: "Meles meles", kingdom: "Animalia"
})
```

✓ 3. External GBIF API Returns Verified Ground Truth — 200 OK · Deterministic

```
{
  "usageKey": 2433857, // GBIF Taxon Key
  "scientificName": "Meles meles (Linnaeus, 1758)",
  "status": "ACCEPTED", "matchType": "EXACT"
}
```

1. Taxonomic Grounding (GBIF Backbone API) — Live Resolution

Never let the LLM guess scientific names, authorship, or taxon keys. Connect the model to the live **GBIF Species API** to verify synonyms, resolve typos, and bind records to official usage keys.

2. Vocabulary Grounding (Darwin Core Quick Reference) — TDWG Standard

Provide tools that strictly enforce valid DwC terms (`decimalLatitude` , not `lat`) and controlled vocabularies (`basisOfRecord: PreservedSpecimen`). Disallow invalid terms.

3. Geospatial & Temporal Grounding — WGS84 / ISO 8601

Use geographic boundary tools (GADM, Nominatim) to detect lat/lon inversion errors and verify that points land inside the stated country. Parse ambiguous dates (`14/05/2023`) into strict `YYYY-MM-DD` .

The Paradigm Shift: From Memory to Tool Orchestration — GROUNDED AGENT

The LLM ceases to be a fallible memory bank. It becomes an **intelligent reasoning coordinator**: it parses natural language, decides which external tools to call, and returns 100% verified scientific data.

The Grounding Rule: Never ask an LLM to *remember* scientific facts. Use the LLM for what it excels at (translating unstructured human messy text into intent), and anchor every fact using **authoritative APIs (GBIF, DwC, WGS84)**.

DEMO: CHATIPT TOOL WORKFLOW →

THE CRITICAL INGREDIENT FOR AI IN SCIENCE

tools, tools, tools...

APIs · Schemas · Validators · Code Interpreters